

THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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ABSTRACT

We investigate how physical scale and population size influence the emergence of complex behavior across open-ended ecological simulations. Rather than optimizing agents for a prescribed benchmark objective, our setting supports unsupervised agents that evolve through reproduction, mutation, and selection while continually reshaping the resources and populations around them. This framing lets us study behavior as a population-level outcome of ecological pressure, not as the performance of a single trained policy on a fixed task. Using modern hardware together with a new multi-agent simulator, we scale both world area and population size beyond prior ecological simulations in this style, reaching more than 60,000 agents, each carrying its own evolved neural network policy. Across the environments studied here, we observe emergent behaviors including long-range resource extraction, vision-guided foraging, and predation under competitive and survival pressures. We analyze how sensing modalities and environmental scale affect whether these behaviors appear, and find that several behaviors become reliable only when worlds and populations are large enough to sustain the relevant ecological conditions; larger scales also make the observed patterns more stable across seeds. Although evolutionary computation and artificial life have long histories, these scaling results indicate that modern accelerators and efficient simulation frameworks can make ecological dynamics a broadly useful instrument for studying machine learning systems under increasingly abundant computation. Experimental code will be released upon acceptance.

1 INTRODUCTION

Since Darwin, the evolutionary emergence of biodiversity and complex behavior has been investigated across many biological sciences, including evolutionary theory (Darwin 1859; Simpson 1944), ecology (MacArthur & Wilson 2001), and genetics (Dobzhansky 1937). Across these areas, models of emergent dynamics range from accounts of speciation and adaptive radiation (Coyne & Orr 2004; Schluter 2000) to mathematical analyses of population genetics (Fisher 1930; Wright 1931).

These disciplines have developed a broad toolkit of mathematical models, laboratory studies, and field observations for explaining how evolution operates in natural systems. Yet the same ecological richness that makes natural systems scientifically valuable also makes them difficult to control, measure, and perturb. As ecosystems grow in spatial extent, population size, and interaction complexity, controlled experiments become harder to run, and interventions in large natural habitats can damage or displace wild populations.

To examine how competitive and environmental pressures can generate complex behavior in a more controllable setting, we study the open-ended evolution of neural network policies in large-scale ecological simulations. Agents in these simulations are not assigned a task reward or external objective; instead, they gather resources to survive and reproduce under the dynamics of the environment. Policy variation enters through mutation when parents pass policies to offspring. With the rapid growth of computational resources available to researchers, this setup lets us probe evolutionary dynamics in populations of simple digital organisms at scales that were previously difficult to access, providing a platform for asking how environmental scale and population size shape behavioral emergence.

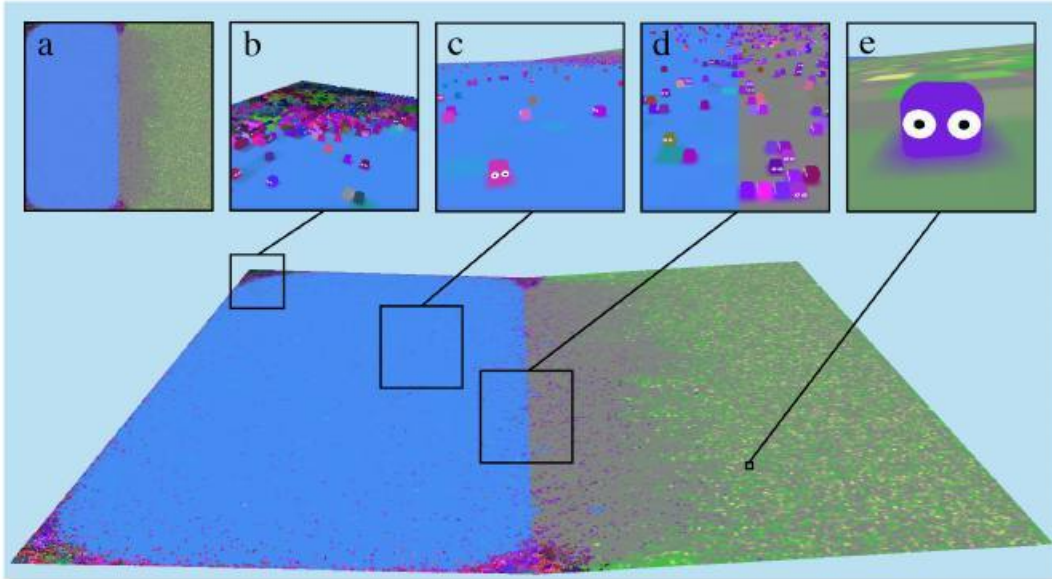


Figure 1: 3D visualization of a 512×512 map in the environment used throughout the paper. Insert (a) gives a top-down view of the full map, (b) shows agents clustered in a corner, (c) shows agents in water, (d) shows agents traveling between land and water, and (e) shows a close-up of one agent gathering resources.

We instantiate this approach in a new JAX-based simulation environment that supports rapid evaluation of large grid worlds. Our largest experiments include more than 60,000 individual agents and physical areas exceeding 1,000,000 grid cells. Within these worlds, agents must navigate, gather food and other resources, and reproduce by accumulating enough resources to create mutated offspring. A 3D visualization of the environment is shown in Figure 1.

Our aim is not only to use simulation as a tool for studying ecological phenomena, but also to evaluate ecology itself as a mechanism for producing machine intelligence. Recent AI results suggest that scale can do more than improve performance; in some settings, it changes the capabilities that systems exhibit (Wei et al., 2022a). Despite this lesson, many embodied-learning studies still operate in relatively small worlds or with limited population sizes. Here we ask what changes when scaling is applied to an entire evolving ecosystem rather than to a single model. Just as reasoning-like abilities can appear in individual models only beyond certain scale thresholds (Wei et al., 2022b), we seek to identify strategies that arise in populations only when environmental size, population density, and ecological structure are sufficiently rich.

Across our experiments, sensor configurations and environmental scale strongly affect the behaviors that emerge and the long-term ecological states that follow. For example, agents equipped with simple compass sensors can evolve long-range resource-gathering expeditions, but this behavior is less reliable in smaller worlds. Agents with vision sensors evolve stronger foraging and attacking behavior than agents without vision, and these differences again become clearer at larger scales. These results show that the scale and complexity enabled by modern accelerators can support a general experimental program for studying how ecological evolution produces rich behavior without explicit objectives.

2 RELATED WORK

Artificial life has drawn on the emergence of biological organization since the beginning of computational research. Conway’s Game of Life (Games 1970) showed how simple local rules can generate population-level structure. Early systems such as Tierra (Ray, 1992) and Avida (Brown) modeled life as self-replicating programs that evolve through mutation. Similarly, Reynolds (1987) used local

motion rules to produce emergent flocking and schooling behaviors, illustrating how macroscopic patterns can arise from local interactions.

Researchers have applied these principles to evolve virtual-agent locomotion morphology (Sims, 1994a,b, Ha, 2018, Gupta et al., 2021, Auerbach et al., 2014, Auerbach & Bongard, 2012, Jeon et al., 2025, Auerbach & Bongard, 2014, Szerlip & Stanley, 2013) and sensing (Pratt et al., 2022, Mugan & MacIver, 2020, Tiwary et al., 2025), while neuroevolution methods have evolved neural network architectures directly (Stanley & Miikkulainen, 2002, 2004, Stanley et al., 2009). In contrast to these approaches, we do not use a reward or objective signal to make agents better adapted to a specified task. Instead, we ask what behaviors emerge from ecological pressure itself and how those behaviors depend on environmental scale and complexity.

Evolution has also shaped the internal dynamics of optimization algorithms. Genetic algorithms (Holland, 1992, Jong, 1975) and evolutionary strategies (Wierstra et al., 2014, Hansen & Ostermeier, 2001, Rechenberg, 1973, Schwefel, 1995, Conti et al., 2018) use variation and selection for optimization, and related methods encourage novelty among discovered solutions (Lehman & Stanley, 2011, Cully et al., 2015). Other works have used these approaches to optimize neural networks for domains such as Atari games and Mujoco locomotion tasks (Salimans et al., 2017, Such et al., 2017, Mania et al., 2018). Although our agents are updated through evolutionary mechanisms, our focus is studying emergent behavior rather than introducing an optimizer or applying an existing optimizer to a fixed objective.

There is also a rich history of biologically inspired work on organisms developing within simulated environments. Early examples include Bedau (1992), Yaeger et al. (1994), Channon & Dampier (1998), and Tisue et al. (2004). Recent works use cellular automata to grow organisms with emergent morphologies and behaviors (Mordvintsev et al., 2020, Robinson, 2021, Heinemann, 2021, Randazzo & Mordvintsev, 2023, Caussan, 2025), Mordatch & Abbeel (2018) and Park et al. (2023) use simulated environments to investigate language and communication emerging between agents. Behavioral emergence through autocurricula (Leibo et al., 2019), where populations of agents provide increasingly challenging learning landscapes for one another, has also been explored in environments with smaller populations than ours (Leibo et al., 2018, Team et al., 2021). Additionally, Wang et al. (2019) and Aki et al. (2024) take a complementary approach by evolving environments optimized for agent learning. Although those works evolve environments rather than agents, they support the broader view that environmental complexity can elicit sophisticated behavior.

Most closely related to our work are efforts that simulate populations of agents interacting within complex environments. Baker et al. (2019) examine a Hide-and-Seek task in which a small number of hider agents learn to use environmental tools to evade seeker agents, with both groups trained via reinforcement learning. Other works similarly train populations for specific tasks: playing video games (Jaderberg et al., 2019), manipulating objects (Petrenko et al., 2021), or capturing agents (Zheng et al., 2017, Yamada et al., 2020). These works demonstrate important forms of emergence, but the emergence is typically bounded by fixed tasks and objectives. The Neural MMO framework (Suarez et al., 2019, 2023) moves toward a more open-ended reinforcement-learning setting by placing agents in a massively multi-agent game and optimizing survival rather than task-specific rewards. However, the population remains capped at a fixed number of agents (experiments examine up to 128 concurrent agents) and lacks generational inheritance, limiting direct study of population dynamics. Hamon et al. (2023) is closer to our setting through variable population sizes, genetically inspired evolution, and generational timelines, though its environment is substantially smaller, less feature-rich, and does not directly study scale. Similarly, Lu et al. (2024) also simulates agents evolving to gather resources. However, this work also operates at a smaller scale (a maximum of 256 agents) and emphasizes tool use through programmable robots. Taken together, prior systems provide many ingredients for ecological emergence; our work combines naturally changing populations, objective-free evolution, richer resource dynamics, and larger spatial scales so that collective ecological patterns can be examined under broader conditions.

Finally, we are also inspired by recent work on environment speed and scale, including optimized libraries that increase computational throughput and the number of agents able to interact in game-like environments (Koyamada et al., 2023, Matthews et al., 2024, Lechner et al., 2023). We build on this progress in computational efficiency by using related JAX-based acceleration techniques to scale ecological environments with evolving populations.

3 METHOD

3.1 ECOLOGICAL GAMES

We formalize interactions between evolving populations and their environment as a multiplayer model that we call an **Ecological Game (EG)**. As in Markov decision processes (MDPs) and stochastic games (SGs) from reinforcement learning and multi-agent reinforcement learning (see Sutton et al. (1998) and Albrecht et al. (2024) for reference), an EG retains Markovian transition dynamics, but it is intended for settings where the population itself changes and no external reward function defines success. Like an SG, an EG contains a set of agents I , a state space S , an initial state distribution ρ_0 , an action space A , and a transition function $\mathcal{T}$ mapping states s_t and a set of actions for each player $a_{t,i}$ to a distribution over subsequent states s_{t+1} . Unlike an SG, it removes the reward function and adds a population function $\Gamma(s_t)$ that records which agents in I are alive and identifies their parents, making births, deaths, and lineages part of the modeled system. Just as MDPs and SGs have partially observable counterparts, POMDPs and POSGs, an EG can also be partially observable, in which case we refer to it as a POEG. A POEG differs from an EG by replacing full-state access with an observation function $O(s_t, i)$ that maps a state and player to that player’s observation. The environment used in this paper is one instance of a POEG, with observation functions determined by the sensor configurations described next.

3.2 ENVIRONMENT

The POEG instance used for our experiments is a large grid world with resources that agents must collect to survive and reproduce. This environment maintains a factored state space $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$ where s_t^m is 2D data associated with the map, such as the terrain and distribution of resources, and s_t^a is data associated with the agent, such as its position, health, and internal resources.

Terrain: Each map contains a rock layer that defines the height of every grid cell. Water is initially placed on cells whose rock value falls below a preset sea level, and during an episode it moves downhill under a discrete flow simulation induced by this height field. The experiments use six global terrain configurations, shown in Figure 2: **Ocean**, **Beach**, **Island**, **Lake**, **Isthmus**, and **Channel**. To separate effects of spatial extent from effects of initial density, we evaluate 64×64 , 128×128 , 256×256 , 512×512 , and 1024×1024 grids while scaling the initial population in proportion to map area. After initialization, population size is no longer fixed; it grows or shrinks according to the ecological dynamics of each run.

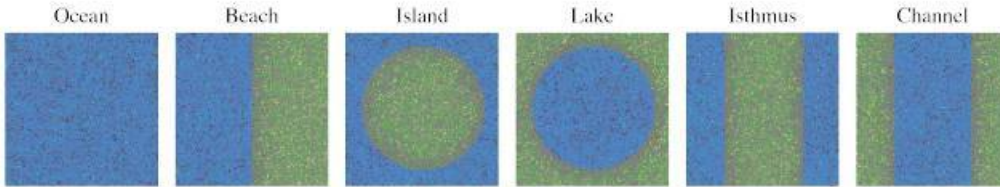


Figure 2: Overhead views of the terrain families used in our experiments at 256×256 grid size. Small dark spots indicate individual agents, and green and yellow indicate concentrations of energy and biomass. Each image was captured 5000 steps into an episode, after most biomass in water had been consumed but before agents had evolved longer-term survival on land.

Resources and Agent Health: Agents must collect **water**, **energy**, and **biomass** to survive and reproduce, making resource acquisition a shared constraint across all terrain types studied here. Biomass and energy are initially distributed randomly throughout the environment, after which individual agents can consume and redistribute them.

Every action consumes a small amount of energy and water. Spent water returns to the landscape and flows downhill toward the nearest body of water, whereas spent energy is removed. Biomass supports reproduction: all agents require a fixed biomass amount to exist, so reproduction succeeds only after an agent has gathered enough excess biomass to allocate to offspring. Total biomass, like total water, is conserved over a simulation, but biomass neither flows downhill nor moves autonomously. Energy is not conserved; it slowly grows on free-standing biomass. These resource

rules create a general multi-resource survival problem rather than a single-food-source collection task. For ablations on resource rules, see Appendix F.2

Agents also maintain health points (HP), which provide a common death mechanism across resource shortages, attacks, and aging. If an agent overspends a critical resource, its HP deteriorates rapidly, although energy and water can be used to recover HP slowly over time. Agents may attack other agents and reduce their targets' HP, and they also receive a small age-dependent damage term that eventually causes death by old age if no other cause intervenes. When HP reaches zero, the resources carried by the dead agent are deposited back into the grid cell where it stood. This rule allows killing to become predation: an attacker that kills another agent can subsequently consume the released resources. For ablations on HP hyperparameters, see Appendix F.8

Sensors: Agents observe the world through four sensor types that let us vary information access without changing the underlying ecological rules. The **internal** sensor reports age, health, and carried resources; the **external** sensor reports resources in the agent's current grid cell; a **compass**, inspired by magnetoreception in birds and other animals, reports global heading as a 4-way one-hot vector; and **vision** returns a 7×7 top-down rendered patch around the agent with three color channels plus a fourth channel encoding local elevation differences. For ablations on vision configurations, see Appendix F.6

Actions: Agents choose from a finite discrete action set organized into five categories. The first is **Rest**, which has no behavioral effect and carries the lowest energy cost. Three **Move** actions rotate clockwise, rotate counterclockwise, or move forward one grid cell. Occupancy is exclusive: if an agent attempts to enter an occupied cell, or if multiple agents attempt to enter the same cell, the corresponding move is canceled. A single **Attack** action kills all agents in the 3×3 square directly in front of the attacker. For ablations on attack strength, see Appendix F.7. Three **Eat** actions correspond to the three resource types. When an agent shares a cell with a resource, the matching eat action transfers a fixed amount of that resource from the environment to the agent's fixed-capacity stomach.

Finally, agents have a **Reproduce** action that implements single-parent reproduction. This action succeeds only when the parent has accumulated enough resources to endow a child. The offspring is placed directly behind the parent, receives a mutated copy of the parent's policy weights, and immediately inherits a fixed quantity of the parent's resources.

To isolate how capabilities affect ecological outcomes, the experiments selectively disable sensors or the attack action while holding the rest of the environment fixed. We use three sensing combinations: internal+external resources only (**R**), internal+external+compass (**RC**), and internal+external+compass+vision (**RCV**). We append **+A** when attacking is enabled; for example, (**RC+A**) denotes agents with resource sensors, a compass, and attack ability.

3.3 POLICIES

Agent policies are small, memoryless MLPs, a deliberately simple model class that keeps the focus on population and environmental scale rather than recurrent state or large architectures. Raw sensor values are flattened, normalized to $[-1, 1]$, and passed through separate linear layers with output dimension 64. The resulting embeddings are summed and processed by a 2-layer MLP with hidden dimension 64 and ReLU nonlinearities. Because the action space is discrete, the final layer has output dimension equal to the number of possible actions. Actions are sampled from a softmax distribution whose temperature evolves along with the network weights. We represent weights using half-precision bfloat16 values to support larger populations and faster runtimes. Depending on the enabled sensors, each agent network contains roughly 10k-25k parameters. For ablations on network size and architecture, see Appendix F.5

For every run, initial policy weights use Kaiming initialization (He et al., 2015), and bias vectors are initialized to zero. When agents are born during an episode, their policies are produced by copying the parent's weights and adding Gaussian noise with standard deviation 3×10^{-2} to all mutable weight values. This mutation rule follows the spirit of prior neuroevolution work (Such et al., 2017). More elaborate evolutionary operators such as crossover could be incorporated, but we use single-parent noisy inheritance to keep reproduction simple and to make the role of environmental scale

easier to interpret. Empirically, the low-precision floating-point representation required a larger mutation rate. We perform an ablation on mutation rates in Appendix F.4

Agents also carry a 3-channel color trait that determines their rendered appearance on the map. This trait mutates in the same manner as the network weights and is visible to other agents through the vision sensor. In principle, color can therefore support distinctions among agent lineages or behavioral types.

3.4 COMPUTATION AND REPRODUCIBILITY

All experiments are implemented in JAX (Bradbury et al. 2018) and executed on Nvidia H100 and A40 GPUs using a state-of-the-art simulation environment. Unless otherwise noted, each run lasts for 2M environment steps or ends earlier if the population goes extinct. Per-seed wall-clock time ranges from under an hour for 64×64 grids to approximately 10 hours for 1024×1024 grids on one H100 GPU. Appendix E provides additional simulator details, scaling measurements, and reproducibility information.

4 EXPERIMENTS

The experiments evaluate how sensory access, spatial scale, and population size shape emergent behavior under the same ecological rules. We keep the datasets, tasks, and evaluation conditions fixed within each comparison, and vary only the capabilities or scales needed to test each question.

4.1 LONG-DISTANCE RESOURCE GATHERING

Our first experiment asks when agents can evolve long-distance excursions from water onto land to recover resources. In terrains with clear directional structure, agents equipped with a compass (RC) develop substantially more effective long-range resource gathering than the baseline (R) agents without a compass. The same comparison also tests scale: this behavior appears more consistently and produces stronger ecological effects in larger environments.

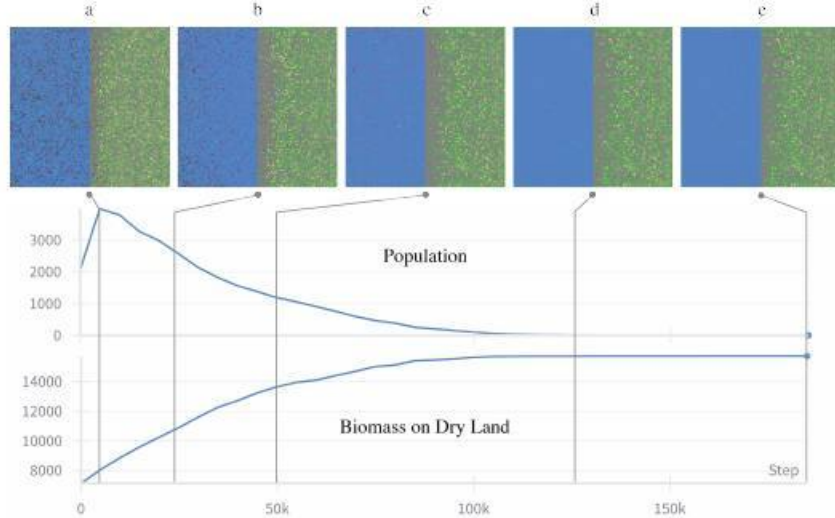


Figure 3: Representative trial with (R) agents in a 256×256 Beach environment. The top chart tracks population over time, and the bottom chart tracks free biomass on dry land.

Figure 3 illustrates the baseline (R) dynamics. After 5000 time steps (a), the initial land agents have died from lack of water, while agents in water have expanded rapidly by consuming abundant initial resources. At this stage, the population has not yet produced policies that reliably search for resources. From this early population peak, some agents occasionally stumble onto land (b). Without a directional cue, they often fail to return to the ocean and die, depositing biomass on

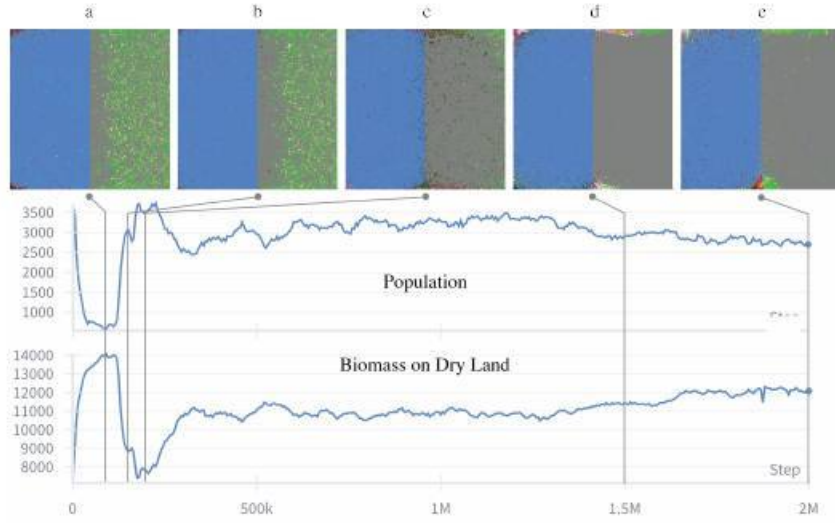


Figure 4: Representative trial with **(RC)** agents in a 256×256 **Beach** environment. The top chart tracks population over time, and the bottom chart tracks free biomass on dry land.

land. By 50,000 time steps (c), this repeated process has nearly doubled the uneaten biomass on land. Moving resources from water to land reduces the biomass available to sustain the aquatic population, causing the population to decline. By 125,000 steps (d), nearly all biomass has been transferred to land and the population is close to extinction; the final agent dies around 185,000 steps (e).

This run shows how a local behavioral limitation can create a population-level ecological shift. Individual agents die because they cannot return to water, but those deaths also redistribute resources in ways that affect the entire population. The same redistribution creates an exploitable niche: any lineage that can travel onto land, consume the accumulated biomass, and return safely to water can convert an ecological collapse into a resource-gathering strategy.

Figure 4 shows the corresponding condition with an internal compass **(RC)**. The early trajectory is similar (a): population decreases while free biomass on land increases. In this run, however, some agents use the compass to drift toward the relative safety of the corners. Around 150,000 steps (b), agents begin to “mine” biomass on land by navigating between land and water. By 200,000 steps (c), these miners have cleared the beach. The population later changes modestly, with agents forming edge clumps (d), but it remains broadly stable through the full 2,000,000 time steps (e).

Table 1: Mining events in the **Beach** world across five world sizes and initial populations, comparing agents with compass access **(RC)** to agents without a compass **(R)**. denotes the number of seeds with a mining event. denotes the number of seeds where the population went extinct before 2M steps. See Table 3 in Appendix A for seed-level data.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	32768		8192		2048		512		128	
(RC)	4/4	0/4	4/4	0/4	4/4	0/4	3/4	1/4	2/4	3/4
(R)	1/4	0/4	0/4	4/4	0/4	4/4	0/4	4/4	0/4	4/4

We quantify this pattern with a full set of **Beach** experiments at grid sizes 1024×1024 , 512×512 , 256×256 , 128×128 , and 64×64 , initialized with 32768, 8192, 2048, 512, and 128 agents respectively. Each condition is run with four random seeds. We mark mining as emerging when free biomass on dry land drops by at least 10% from its historical maximum. We also record the mining

onset time, defined by the local maximum in land biomass, and the resulting percentage biomass drop. Table 1 summarizes mining and extinction events, and Table 3 in Appendix A reports seed-level details. Across these matched conditions, mining appears more consistently for (RC) agents than for (R) agents. It also becomes more reliable with scale, occurring in every seed at 256×256 and larger. At smaller scales, populations are less stable and more likely to go extinct early.

To characterize timing rather than only final outcomes, we run additional (R) and (RC) seeds at 512×512 world size (32 seeds) and 128×128 world size (64 seeds). We then use a Kaplan-Meier estimator to measure the probability of extinction and mining over simulator time. Extinction is rarer and occurs later in larger worlds, while mining is more frequent and occurs sooner at smaller grid sizes. We also fit a Cox proportional hazards model to compare mining-discovery and extinction timing across world sizes and compass conditions. This analysis likewise finds that larger world size substantially lowers extinction rate and enables much faster mining discovery than smaller worlds; enabling the compass has a similar effect. Full details can be found in Appendix B.

Table 2: Mining and extinction events across the **Island**, **Lake**, **Isthmus**, and **Channel** terrains at 512×512 , comparing agents with compass access (RC) to agents without a compass (R).  denotes the number of seeds with a mining event.  denotes the number of seeds where the population went extinct before 2M steps. See Table 4 in Appendix A for seed-level data.

	Island		Lake		Isthmus		Channel	
								
(RC)	4/4	0/4	2/4	2/4	4/4	0/4	3/4	1/4
(R)	3/4	1/4	2/4	4/4	1/4	3/4	0/4	4/4

We next test whether the compass effect extends beyond the **Beach** layout by running **Island**, **Lake**, **Isthmus**, and **Channel** terrains at 512×512 with (R) and (RC) agents. Table 2 reports mining and extinction events, and Table 4 in Appendix A gives run-level details. As in the beach setting, compass access yields earlier and more consistent mining, along with more stable populations and lower extinction rates. (RC) agents adapt more readily to the island and isthmus than to the lake and channel, where extinction sometimes occurs. We hypothesize that the island and isthmus provide a simpler directional rule because walking straight east or west reliably returns an agent to water, whereas the lake and channel do not provide an equally direct rule. The (R) agent class does not reliably adapt to any of these terrains, though it performs least poorly on the island. See Figure 12 in Appendix D for a visual example of mining in the island terrain.

4.2 VISUAL FORAGING AND PREDATION

We also use the **Ocean** environment to test whether visual observations support more effective resource discovery and predation. The comparison holds resources, compass sensing, and attack ability fixed while adding a 7×7 visual sensor: (RCV+A) agents are compared against (RC+A) agents with resource and compass sensors only.

Figure 5 reports population, biomass utilization, movement actions, and eat actions for 512×512 and 128×128 oceans. Additional grid sizes appear in Figure 9 in Appendix C and Table 5 in Appendix A summarizes the same measurements. Biomass utilization is the percentage of total environmental biomass currently held inside agents. Runs with vision sensors (red lines) spend less time moving and more time eating. They also sustain larger populations and higher biomass utilization. These patterns indicate that agents can adapt visual information into more efficient foraging behavior. Across environmental scales, between-run variance decreases substantially in larger worlds. Runs with vision also show more stable dynamics over time, with less temporal variation in these metrics.

Figure 6 reports attacks per agent and homicides per attack for the same runs. Agents with vision attack relatively infrequently but maintain an approximately 75% successful kill rate. In large grids, agents without vision sometimes also reach high kill rates, but their dynamics are substantially less stable. Several confounders make the cause of this effect difficult to isolate. Visual inspection (see Figure 13 in Appendix D) suggests that high blind-agent kill rates sometimes occur when clusters accumulate in one location, whereas such clusters almost never form in visual-agent runs. This

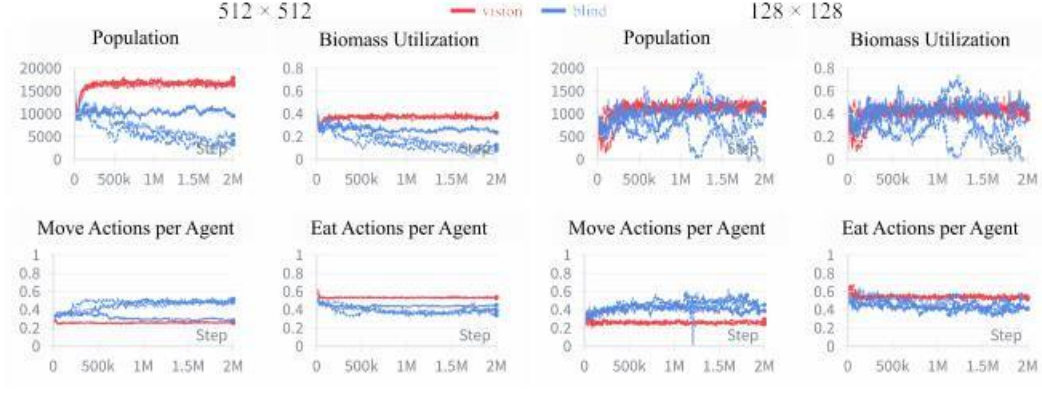


Figure 5: Population, biomass utilization, move actions per agent, and eat actions per agent in 512×512 and 128×128 **Ocean** worlds. Biomass utilization, move actions per agent, and eat actions per agent are averaged across agents at each time step. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). Vision supports larger and more stable populations, higher biomass utilization, less movement, and more frequent effective eating. At larger scales these dynamics have lower variance across runs, indicating that visual sensing provides a stronger foraging advantage in larger environments. See Figure 9 in Appendix C for a comparison of grid sizes 1024×1024 and 256×256 .

pattern is also consistent with visual agents adapting to avoid one another. In smaller environments, the gap between visual and blind agents is less pronounced, but visual-agent populations remain much more stable.

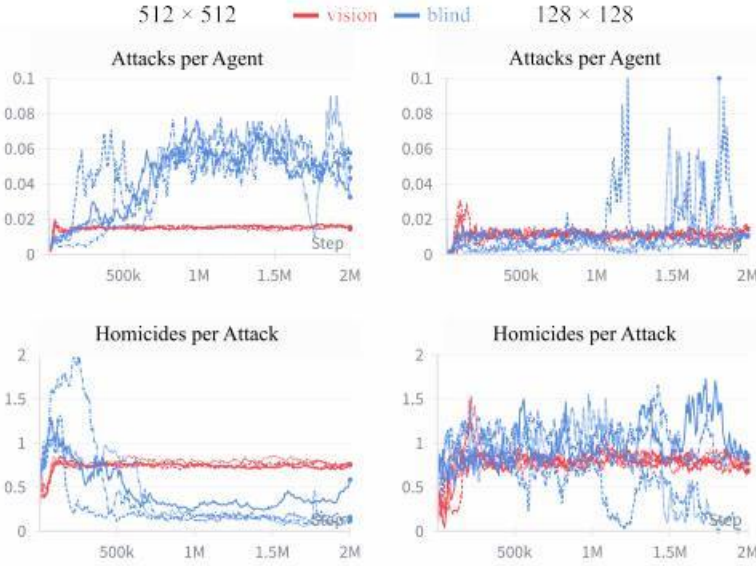


Figure 6: Attacks per agent and homicides per attack, averaged across agents, in 512×512 and 128×128 **Ocean** worlds. Each plot shows four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). Vision leads agents to attack less often but with higher precision, consistent with specialized predation behavior. The pattern is stronger at larger scales. See Figure 10 in Appendix C for a comparison of grid sizes 1024×1024 and 256×256 .

4.3 ABLATIONS

Appendix F broadens the robustness checks by varying initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7),

and agent HP (F.8). These ablations measure how sensitive the observed emergent behaviors are to simulator configuration. Appendix F.3 further tests whether the dynamics in larger environments reflect more than simple self-averaging across larger populations.

5 CONCLUSION

This study supports open-ended ecological simulation as a general paradigm for investigating how intelligent behavior can emerge from survival, reproduction, and environmental feedback. Across worlds that vary in scale, terrain structure, sensory access, and interaction rules, we find that environmental extent and sensory richness both contribute substantially to the development of effective sensing and decision-making policies. Together, these results suggest that ecological and evolutionary pressures, when paired with sufficiently rich environments, can provide a broadly applicable route for developing and analyzing increasingly capable artificial agents. Our experiments also show that large populations can evolve nontrivial behaviors on a practical hardware budget without explicit task objectives. We hope these findings encourage further work using ecological dynamics as a mechanism for both generating and understanding artificial intelligence.

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A ADDITIONAL DATA

Table 3: Mining events in the **Beach** world across five scales, comparing populations with compass access (**RC**) to populations without a compass (**R**).  denotes whether a run had a mining event.  denotes the percent decrease in free biomass on dry land during the mining event, and  denotes the approximate timestep at which free dry biomass reached the local maximum indicating the start of the mining event. For runs with multiple mining events, we report the first one.  denotes the timestep at which the run went extinct, or n/a if it lasted for 2M steps.

	1024x1024				512x512				256x256				128x128				64x64			
	32768				8192				2048				512				128			
Compass On (RC)																				
Seed																				
0	✓	35k	11.1	n/a	✓	105k	33.	n/a	✓	145k	49.3	n/a	✓	105k	72.5	n/a	x			55
1	✓	90k	16.2	n/a	✓	50k	17.2	n/a	✓	50k	46.6	n/a	x		140	✓	70k	85.3	675	
2	✓	40k	15.	n/a	✓	30k	14.3	n/a	✓	55k	34.7	n/a	✓	50k	50.7	n/a	✓	10k	78.	n/a
3	✓	60k	15.	n/a	✓	45k	15.6	n/a	✓	90k	47.4	n/a	✓	20k	52.	n/a	x			135
Avg	4/4	34k	14.3	0/4	4/4	58k	20.	0/4	4/4	85k	44.5	0/4	3/4	58k	58.4	1/4	2/4	40k	81.7	3/4
Compass Off (R)																				
Seed																				
0	x			n/a	x			945k	x			185k	x			65k	x			45k
1	✓	275k	12.4	n/a	x			1085k	x			110k	x			55k	x			40
2	x			n/a	x			1070k	x			165k	x			55k	x			25
3	x			n/a	x			280k	x			125k	x			90k	x			40
Avg	1/4	275k	12.4	0/4	0/4			n/a	0/4			4/4	0/4			4/4	0/4			4/4

Table 4: Mining events across the **Island**, **Lake**, **Isthmus**, and **Channel** terrains, comparing (RC) agents to (R) agents. Each run uses a 512×512 grid with an initial population of 8192.

	Island				Lake				Isthmus				Channel			
Compass On (RC)																
Seed																
0	✓	40k	94.3	n/a	x			45k	✓	45k	33.	n/a	✓	50k	27.4	n/a
1	✓	30k	94.3	n/a	✓	30k	22.5	n/a	✓	55k	37.3	n/a	✓	80k	30.8	n/a
2	✓	30k	94.8	n/a	x			45k	✓	40k	28.1	n/a	✓	50k	41.	n/a
3	✓	70k	95.1	n/a	✓	35k	25.3	n/a	✓	30k	41.5	n/a	x			80k
Avg	4/4	43k	94.6	0/4	2/4	33k	23.9	2/4	4/4	43k	35.	0/4	3/4	60k	33.1	1/4
Compass Off (R)																
Seed																
0	x			185k	x			125k	x			145k	x			100k
1	✓	75k	30.9	n/a	x			130k	x			260k	x			105k
2	✓	85k	47.	n/a	✓	75k	23.4	1400k	✓	95k	18.8	n/a	x			105k
3	✓	115k	39.5	n/a	✓	70k	24.3	1250k	x			140k	x			100k
Avg	3/4	92k	39.1	1/4	2/4	73k	23.9	4/4	1/4	95k	18.8	3/4	0/4			4/4

Table 5: Population and foraging measurements for Vision and Blind agents in **Ocean** worlds across scales. denotes population size. denotes biomass utilization, the percentage of total biomass currently inside agents' stomachs. denotes move actions per agent, the fraction of time an agent selected a move action averaged across agents. Similarly, denotes the fraction of agent actions that selected an eat action. Reported , , , and values are averages over all steps for each seed.

	1024x1024				512x512				256x256				128x128			
	32768				8192				2048				512			
Vision (RCV+A)																
Seed																
0	58354	.396	.402	.426	16413	.369	.257	.532	4195	.382	.255	.536	1126	.426	.261	.541
1	65107	.365	.257	.529	16200	.364	.256	.533	4142	.377	.258	.536	1103	.415	.262	.537
2	64735	.363	.257	.530	16560	.372	.257	.533	4269	.388	.260	.536	1041	.392	.254	.542
3	64595	.362	.256	.532	16206	.365	.255	.531	4249	.388	.259	.533	1100	.414	.257	.538
Avg	63198	.371	.293	.504	16345	.368	.256	.532	4213	.384	.258	.535	1092	.412	.259	.539
Blind (RC+A)																
Seed																
0	22734	.143	.446	.372	6549	.173	.442	.400	2036	.209	.433	.461	545	.219	.465	.427
1	23670	.151	.426	.388	6939	.180	.438	.396	3107	.321	.390	.416	1082	.444	.416	.460
2	24745	.159	.358	.426	5448	.136	.474	.374	1587	.163	.440	.433	979	.421	.404	.436
3	20004	.130	.469	.359	10354	.265	.311	.439	3164	.324	.321	.450	788	.317	.431	.477
Avg	22788	.146	.425	.386	7323	.189	.416	.402	2473	.254	.396	.440	848	.351	.429	.450

Table 6: Predation measurements for Vision and Blind agents in **Ocean** worlds across scales.  denotes attacks per agent, the fraction of time an agent selected the attack action averaged across agents.  denotes homicides per attack, the number of agents killed per attack action. Reported  and  values are averages over all steps for each seed. Note the 0.562 outlier for  in seed 3 of the 128x128 (**RC+A**) run; this seed went extinct before 2M steps, causing attacks per agent to spike as population size dwindled.

	1024x1024		512x512		256x256		128x128	
	32768		8192		2048		512	
Vision (RCV+A)								
Seed								
0	.011	.776	.015	.731	.013	.784	.010	.788
1	.016	.733	.015	.741	.014	.780	.011	.793
2	.016	.707	.015	.737	.014	.750	.012	.771
3	.016	.712	.015	.780	.013	.759	.011	.765
Avg	.015	.732	.015	.747	.014	.758	.011	.779
Blind (RC+A)								
Seed								
0	.070	.218	.042	.437	.005	1.074	.016	.558
1	.054	.300	.044	.383	.016	.871	.006	1.032
2	.076	.236	.048	.238	.031	.383	.010	.979
3	.059	.203	.044	.456	.033	.446	.562	.841
Avg	.065	.239	.045	.378	.021	.693	.148	.853

B DEMOGRAPHY

We compare extinction and mining-emergence rates as functions of world size and access to global direction through the compass sensor. Extinction is defined as the population reaching zero individuals, and mining is defined as in the earlier experiments: a 10% drop in free biomass on land.

Runs are grouped by both world size and compass access. We simulated 64 runs each for 128×128 without a compass and 128×128 with a compass, and 32 runs each for 512×512 without a compass and 512×512 with a compass. For each group, we fit a non-parametric Kaplan-Meier estimator to the time-to-extinction distribution, treating runs that did not go extinct within 2M steps as right-censored. Figures 7 and 8 show that populations tend to survive longer and develop mining sooner when worlds are larger or agents have compass information.

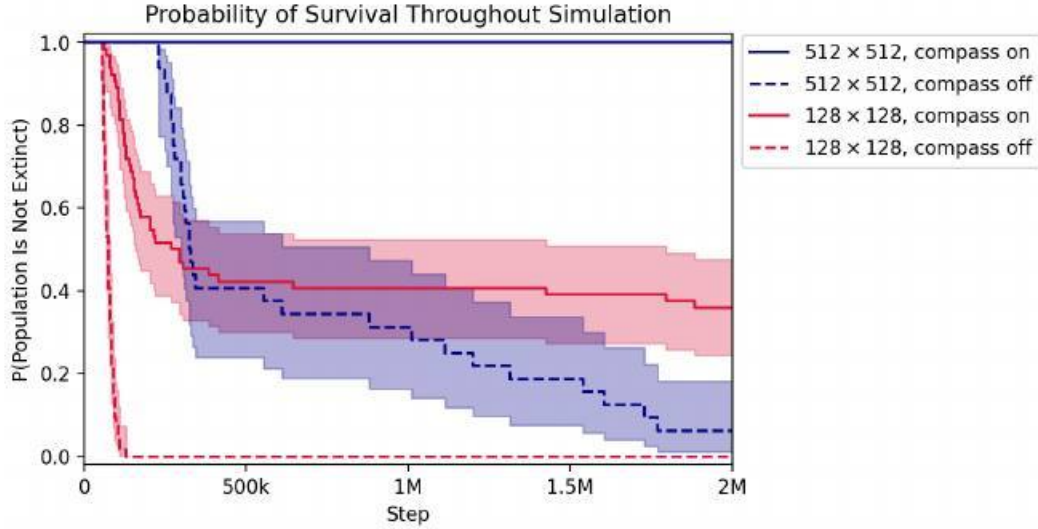


Figure 7: Probability that the population has not gone extinct as a function of world size, compass access, and timestep. A separate non-parametric Kaplan-Meier estimator is fit for each run group (e.g., 512×512 , compass on). Shaded regions show 95% confidence intervals. Each 512×512 group uses 32 runs; each 128×128 group uses 64 runs.

We compare emergence and extinction timing with a Cox proportional hazards model, a semi-parametric survival model that relates covariates to event hazard rates (Table 7). Runs without an emergence or extinction event by 2M steps are treated as censored. For extinction, both larger world size and compass use substantially reduce hazard; the extinction rate in 128×128 worlds is about 25 times the corresponding rate in 512×512 worlds ($HR=0.04$). Compass access produces a similar-magnitude effect ($HR=0.03$). For mining emergence, larger worlds produce mining much more rapidly ($HR=6.07$), and compass access accelerates emergence strongly ($HR=34.59$). The models discriminate well between groups (concordance 0.80 for extinction, 0.85 for mining). These results are consistent with larger worlds sustaining larger populations, lowering the chance of reaching the absorbing zero-population boundary while also exploring a larger phenotype space through more mutation events than smaller worlds.

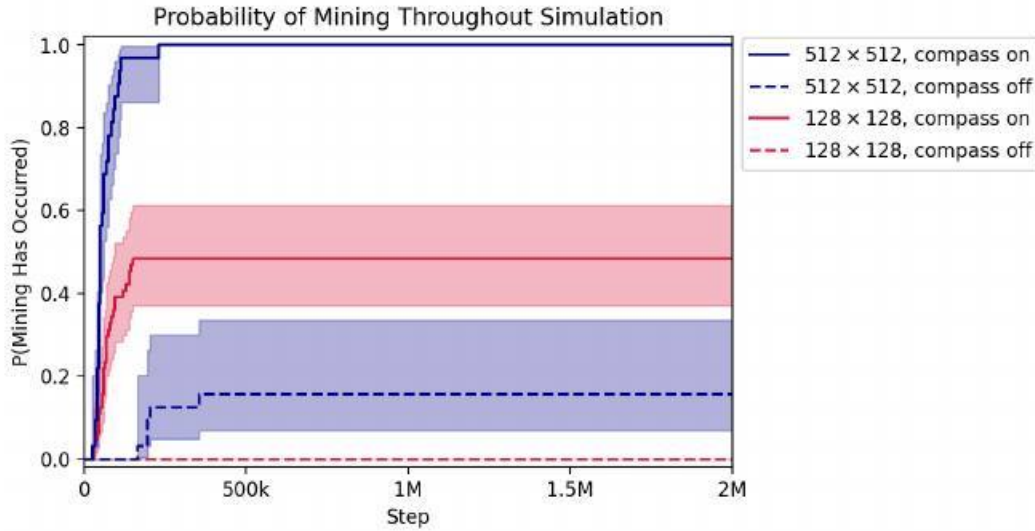


Figure 8: Probability that mining has emerged in the population as a function of world size, compass access, and timestep. A separate non-parametric Kaplan-Meier estimator is fit for each run group (e.g., 512×512 , compass on). Shaded regions show 95% confidence intervals. Each 512×512 group uses 32 runs; each 128×128 group uses 64 runs.

Table 7: Cox proportional hazards models for extinction and emergence events using the 128×128 , compass-off condition as the baseline. Hazard ratios (HR) below 1 indicate reduced hazard and longer time to event; for example, extinction $HR = 0.04$ for 512×512 means extinction occurs at 0.04 times the 128×128 rate. The analysis includes 192 runs: 32 compass-on and 32 compass-off seeds in the 512×512 world, plus 64 compass-on and 64 compass-off seeds in the 128×128 world.

Covariate	Extinction			Mining		
	HR	95% CI	<i>p</i> -value	HR	95% CI	<i>p</i> -value
Size 512×512	0.04	0.02–0.07	< 0.005	6.07	3.61–10.20	< 0.005
Compass On	0.03	0.01–0.05	< 0.005	34.59	13.51–88.57	< 0.005
Concordance	0.80			0.85		
Events / Total	135 / 192			68 / 192		

C ADDITIONAL FORAGING AND PREDATION RESULTS

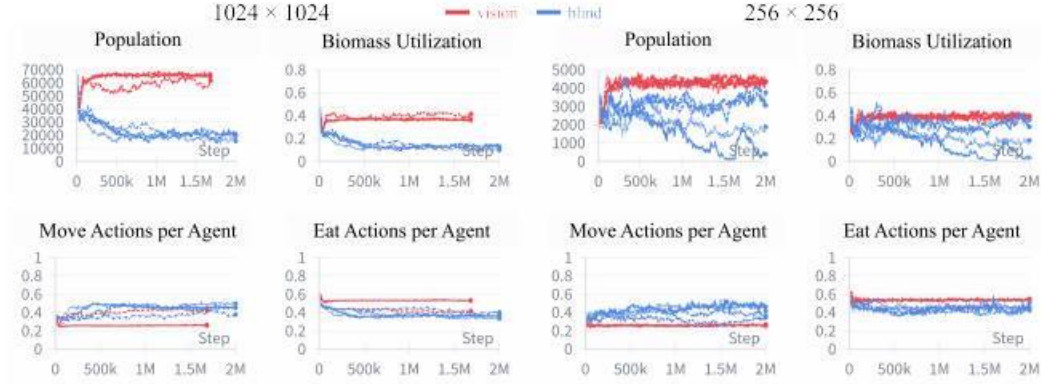


Figure 9: Foraging behavior in the **Ocean** world across scales. Enabling vision (red) yields more stable populations and greater biomass utilization through more frequent effective eating and less movement. Larger scales reduce variance across runs. Due to GPU-hour limits, the vision seeds at 1024x1024 ran for 1.7M rather than 2M steps.

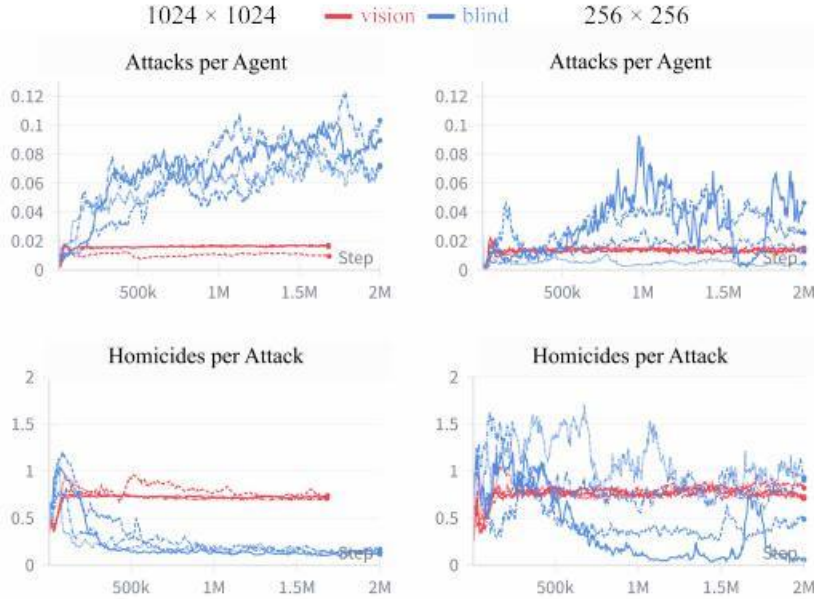


Figure 10: Predation behavior in the **Ocean** world across scales. With vision (red), agents attack less frequently but with much higher precision, consistent with specialized predation in addition to foraging capabilities (Figure 9). The effect is stronger at larger scales, while blind agents (blue) show higher behavioral variance in smaller worlds.

D ADDITIONAL EXPERIMENT VISUALIZATIONS

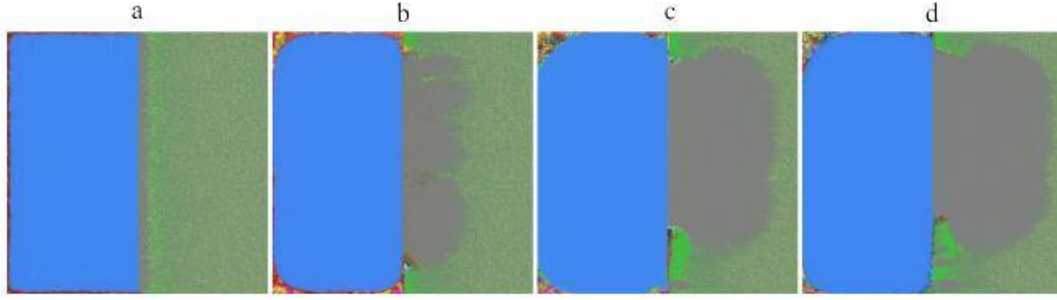


Figure 11: Larger environments support more spatially varied ecologies and distinctive visual patterns. In this 1024×1024 **Beach** world, (RC) agents initially died at the shore (a), transferring biomass to dry land. They then developed mining behavior and cleared biomass from the beach (b). When biomass re-accumulated at the beach ends (c), subpopulations re-developed mining behavior, traveling inland along these smaller biomass beaches to extract resources (d).

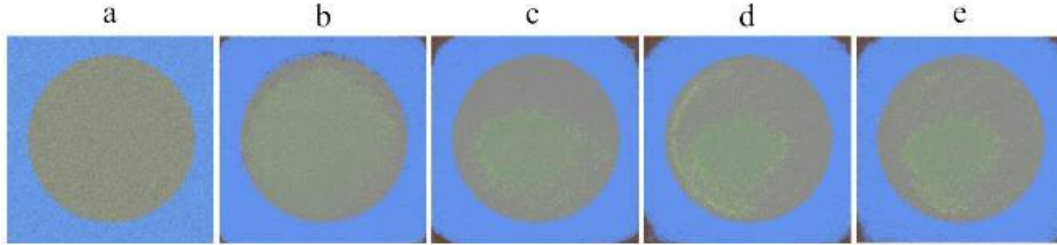


Figure 12: Mining behavior extends to varied terrain structure. In this 512×512 **Island** world with (RC) agents, different shoreline regions are mined during the run (b) (c). When mining becomes difficult, agents die before returning to water and transfer biomass back to parts of the island (d). Agents later adapt to re-mine this accumulated wall of biomass, clearing it from the shore (e).

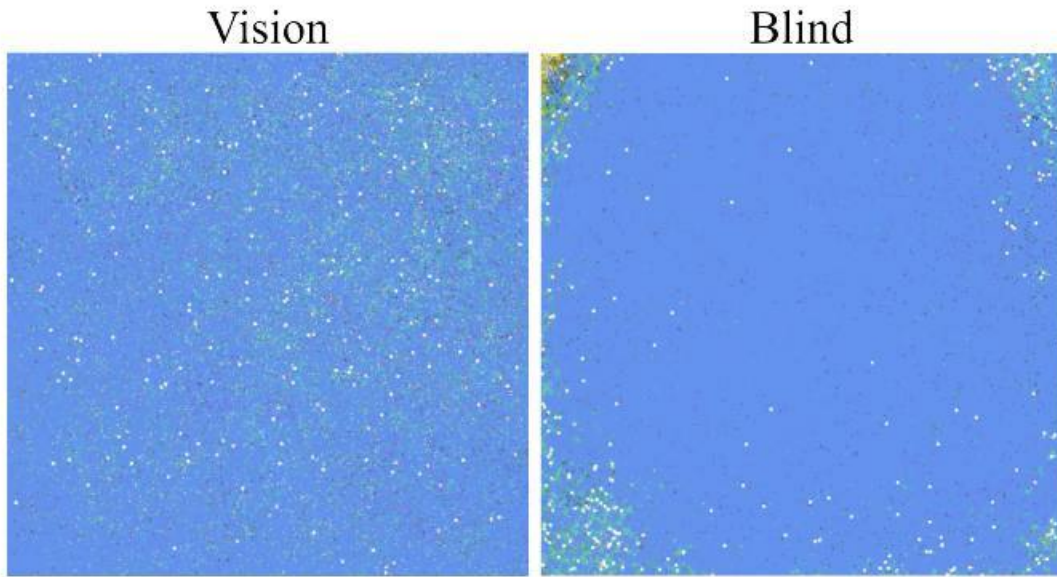


Figure 13: Representative attacking-enabled runs in a 512×512 **Ocean** world: vision (RCV+A) agents are shown on the left, and blind (RC+A) agents on the right. White squares mark locations where agents are attacking. With vision, agents adapt to locate free resources throughout the ocean and prey more effectively on other agents (see Figures 5 and 6). Blind agents explore for resources much less and instead adapt policies that cluster in the world corners.

E DETAILS ON COMPUTATION AND REPRODUCIBILITY

We report simulation-design, runtime, and reproducibility details that contextualize the scale of the experiments.

Simulator design. All experiments use a JAX-based, GPU-accelerated simulator designed for large multi-agent ecosystems with open-ended population sizes. Our simulator computes every part of the environment step on the GPU, including terrain dynamics, resource flows, agent updates, and measurement, so that policy evaluation and simulation share a single JAX compilation pipeline. The simulator exposes a minimal population-level interface: given current map state, agent states, and heritable traits, a population algorithm supplies actions and updated traits and handles births, deaths, and environmental transitions. This separation makes the environment implementation reusable across evolutionary algorithms, reinforcement learning methods, and other population-update rules. All operations are implemented as pure JAX functions combined with vectorization and just-in-time compilation, enabling fusion into a small number of GPU kernels. This yields the throughput required for our scaling experiments, with runtime measurements for a single H100 GPU given in Table 8.

The simulator also includes tooling that supports analysis across experimental settings, including a measurement API for aggregate statistics such as biomass utilization, movement and eating rates, attack and homicide rates, and environment-level parameters at each step, which we use directly in our figures and tables. In addition, an interactive 3D viewer renders terrain, water, resources, and agents from generated reports, and supports inspection of individual agents (Figure 1). These tools were critical for discovering and validating qualitative phenomena such as long-range resource mining and predation patterns that are not obvious from scalar metrics alone.

From a high-performance-computing perspective, the simulator occupies a complementary point relative to existing frameworks. General-purpose agent-based frameworks such as MASON (Luke et al., 2005), FLAME-GPU (Richmond et al., 2024), or Griddly (Bamford et al., 2020) offer high configurability but require substantial environment engineering and do not directly target the open-ended, evolving populations at the scales we study. Conversely, GPU-native multi-agent RL environments emphasize fixed numbers of agents and task-specific reward structures. We combine GPU-native execution, explicit support for births and deaths with mutable traits, and configurable resource and terrain dynamics, enabling open-ended populations of over 10^5 agents while remaining interoperable with diverse population algorithms. This combination yields a state-of-the-art platform for scaling ecological experiments of the kind studied here and provides a foundation for broader work on emergent behavior in large populations. Our simulator will be released upon acceptance.

Runtime data. Table 8 reports runtime across grid sizes and maximum agent populations. Each run was conducted on a single H100 GPU for a maximum of 2M steps. Note that while this table shows run-times for (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ($\sim 22k$ parameters total), the runtime varies with the size of the agent networks, the available sensors, and the level of logging.

Our largest runs at the 1024×1024 grid size take ~ 10 hours in the **Beach** terrain and ~ 14 hours in the **Ocean** terrain on one H100, with memory usage of roughly 60GB. **Ocean** runs are slightly slower due to the enabled agent attack action and vision sensing.

Reproducibility. Because the environments are large and randomized, precise trajectories can differ substantially from run to run. As a result, we report the results of four different random seeds for each experiment. Unfortunately, due to nondeterminism in some low-level GPU operations, we find that repeating an experiment with the same random seed does not reliably produce the same results. In particular, modern GPUs may execute parallel floating-point operations such as summations in different orders across runs and fuse operations in ways that change rounding behavior. Very small differences arising from finite-precision arithmetic can accumulate over many steps into noticeable drift in the simulation trajectory. The use of 16-bit weights can coarsen these rounding effects, but is not the primary source. Despite trajectory-level variation, the high-level patterns reported in the paper are clear when multiple seeds are aggregated.

Table 8: Runtime measurements for **Beach** and **Ocean** worlds across grid sizes and maximum agent populations. Each **(RC)** agent has a 22,465-parameter neural network, and each **(RCV + A)** agent has a 22,529-parameter neural network. We report simulation steps per second (hz) and extrapolated wall-clock runtime in hours for 2M steps. Effective runtime is lower for seeds where populations go extinct before 2M steps.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	131072		32768		16384		8192		4096	
Environment	hz	runtime	hz	runtime	hz	runtime	hz	runtime	hz	runtime
Beach (RC)	55	10.1 hrs	190	2.9 hrs	315	1.8 hrs	535	1.0 hrs	750	0.7 hrs
Ocean (RCV + A)	40	13.9 hrs	142	3.9 hrs	235	2.4 hrs	415	1.3 hrs	750	0.7 hrs

F ABLATIONS

We present ablations that test how broadly the main behaviors persist across environment parameters, agent configurations, and implementation choices.

F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We first vary initial population size in a 256×256 **Beach** world to test whether mining consistency and population stability depend on the starting number of agents rather than the ecological regime reached after initialization. Reported results are for **(RC)** populations. Agents without a compass **(R)** had 0% mining and 100% extinction across all initial population sizes except 8192, the largest tested setting, where they had 25% mining and 75% extinction. Mining and extinction frequencies need not sum to 100% because a population can mine and later go extinct, or survive for the full 2M steps without developing mining.

Figure 14 shows that initial population size has little effect on mining consistency or extinction rate in this setting: populations initialized with only 128 agents evolve mining as often as those initialized with 8192 agents, and extinction rates remain broadly stable across the sweep.

Figure 15 helps explain why the initial condition has limited effect. Regardless of initial population size, runs rapidly approach a similar population level within the first 50,000 simulation steps. This suggests that the environment induces an implicit carrying capacity through the fixed resources available in the system: populations below that capacity grow toward it, while populations above it are constrained downward. After this early correction, effective population sizes are similar, so mining and extinction frequencies are largely unchanged. Larger initial populations do add some resources because agents carry biomass that is deposited on death, and therefore they reach higher maximum populations on average (Table 9); however, this effect is not large enough to change mining or extinction frequency in this setting. Although early populations converge, they need not remain similar throughout a run, because behaviors that emerge at different times can later reshape resource distributions in different ways.

F.2 VARYING RESOURCE RULES

We next vary two resource hyperparameters—biomass photosynthesis rate and initial biomass site density—to test whether the main mining result depends on a narrow resource setting. The setup matches Appendix F.1: initial population size is fixed at 2048, and agents have compass access **(RC)**.

The biomass photosynthesis rate controls how quickly landscape biomass creates energy over time, while initial biomass site density sets the fraction of grid cells containing biomass at the start of simulation. All main-text experiments (Section 4) use a biomass photosynthesis rate of 0.01 and an initial biomass site density of 1/4.

Unlike initial population size (Figure 14), resource rules substantially affect mining and extinction consistency, as shown in Figures 16 and 17.

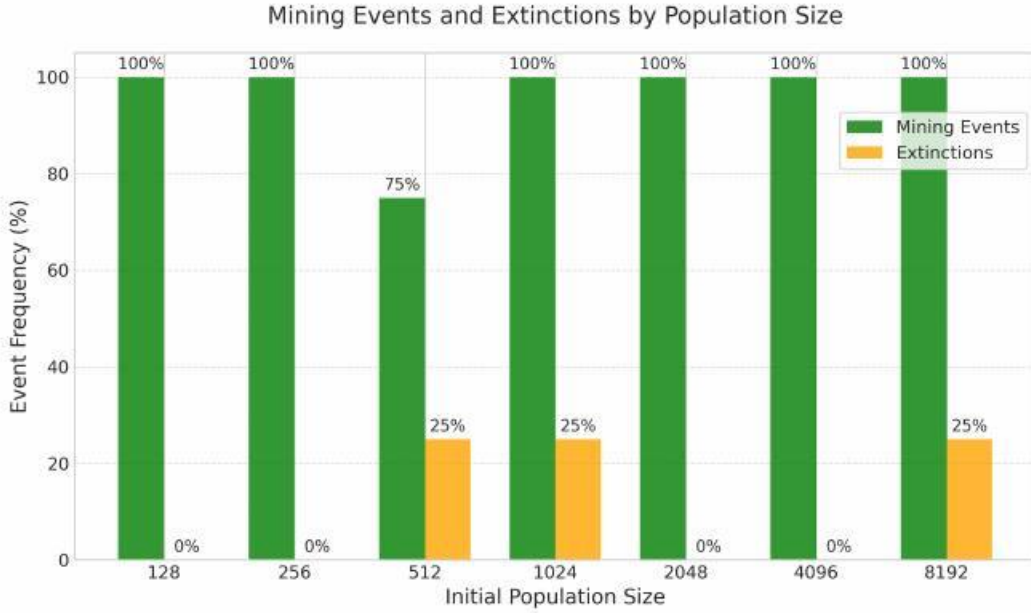


Figure 14: Frequency of mining and extinction events in a 256×256 **Beach** world across initial sizes of (**RC**) agent populations. Frequencies are computed over four random seeds per initial population size; 25% Extinction Event Frequency means that 1/4 seeds reached population 0 before 2M simulation steps.

Table 9: Maximum population size reached after step 0 for each starting population size, reported as averages and standard deviations over four random seeds. Larger initial populations reach slightly higher maxima and show greater seed variance, but the maximum population remains broadly robust to initialization: an initial size of 128, only 1.6% of 8192, reaches 78.9% of the maximum reached by 8192 (3178 vs. 4026).

Initial Population Size	Max Population Size After Step 0
128	3178 ± 217
256	3056 ± 61
512	3451 ± 437
1024	3613 ± 381
2048	3786 ± 431
4096	3689 ± 437
8192	4026 ± 642

Larger biomass photosynthesis rates or initial biomass site densities increase available resources and therefore raise the environment’s effective carrying capacity. Larger sustained populations create more opportunities for mutations that support mining while reducing extinction risk. The data are consistent with this mechanism: omitting extinct seeds, runs with initial biomass site density 1/16 reached an average maximum population of 1608 after step 0, compared with 2466 for density 1/8, 3786 for density 1/4, and 5353 for density 1/2.

F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

One possible explanation for the greater stability of mining in large environments (Table 1) is self-averaging rather than a change in ecological dynamics. By self-averaging, we mean the statistical-physics intuition that, as system size grows, fluctuations in macroscopic quantities shrink roughly as $1/\sqrt{N}$, where N is the system size, such that a single large system behaves like the average of

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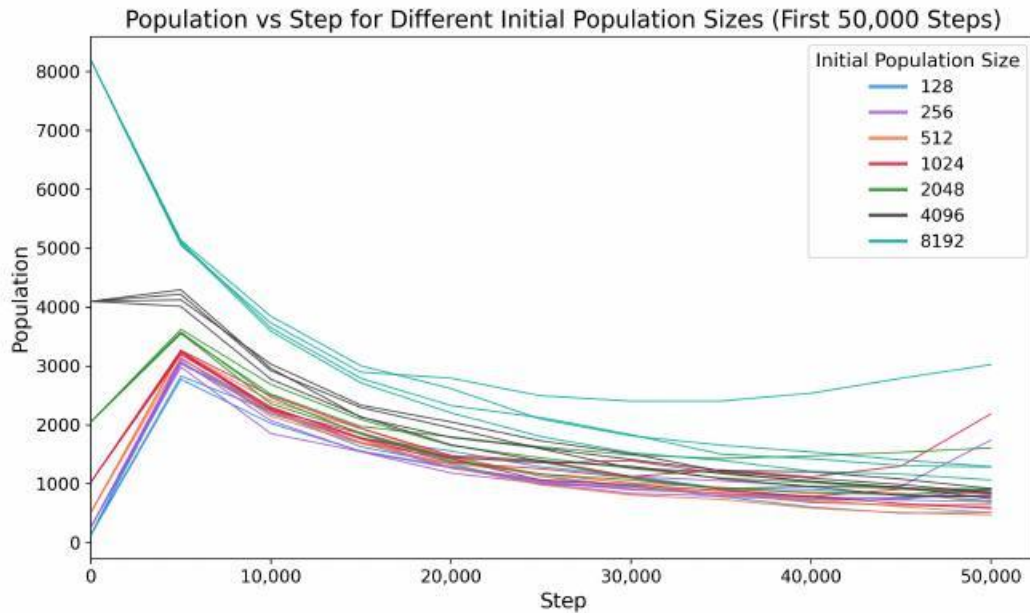


Figure 15: Population size during the first 50,000 time steps in the 256×256 **Beach** with (RC) agents, using four seeds per initial population size. The environment induces an implicit resource-based carrying capacity, and all initial population sizes approach this regime early in the run. Larger initial populations add biomass that is deposited on agent death, raising the effective carrying capacity slightly.

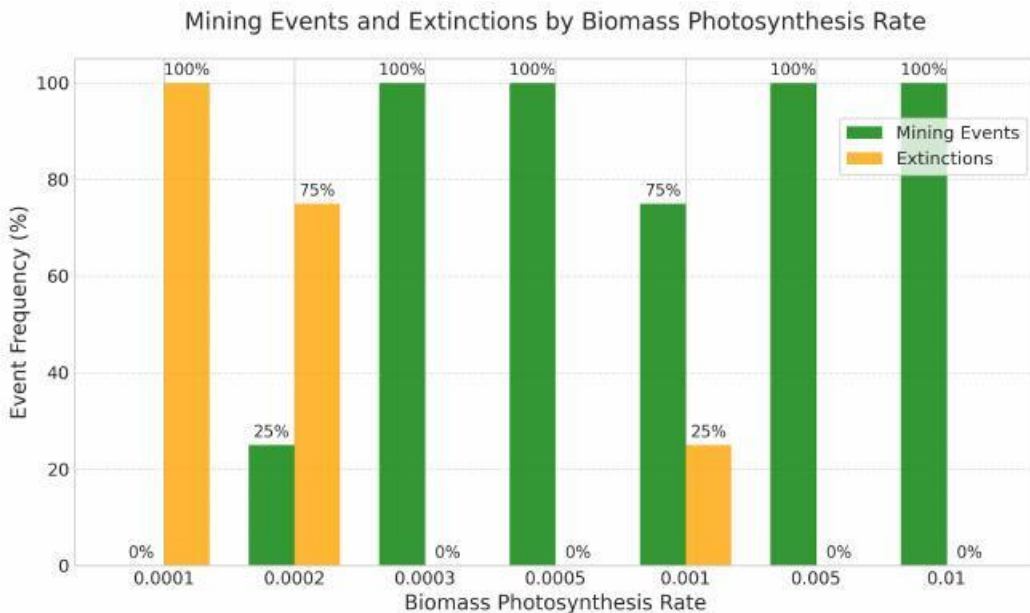


Figure 16: Frequency of mining and extinction events in a 256×256 **Beach** world across biomass photosynthesis rates. Frequencies are computed over four random seeds per setting. Mining emerges consistently when biomass photosynthesis rate is 0.0003 or greater.

smaller systems of the same total size. Under this view, a 512×512 **Beach** world might simply be acting like many loosely coupled 128×128 worlds run in parallel. If that were the case, then our

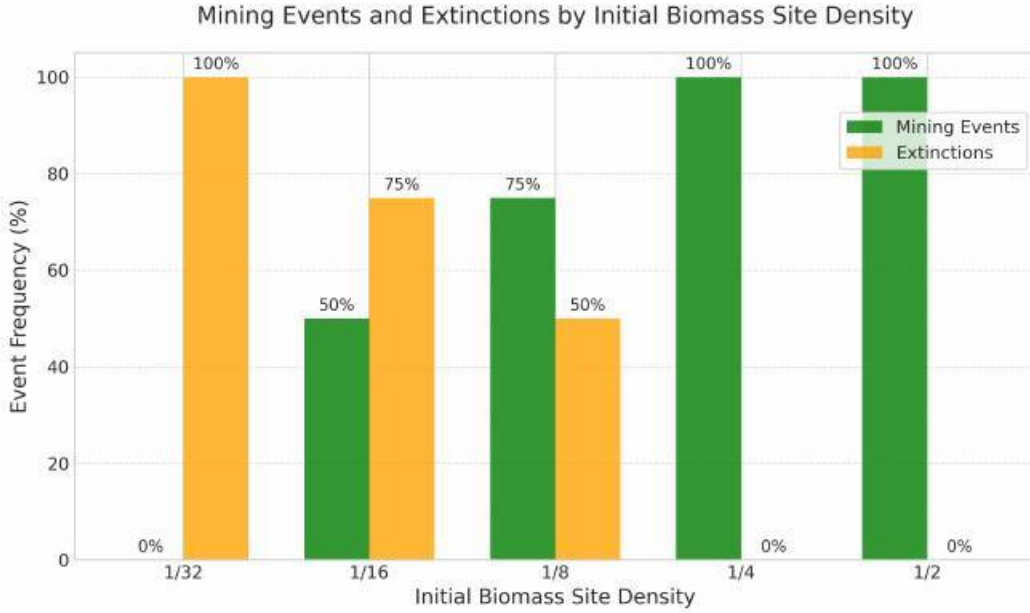


Figure 17: Frequency of mining and extinction events in a 256×256 **Beach** world across initial biomass site densities. Frequencies are computed over four random seeds per setting. Greater initial biomass site density produces more consistent mining discovery and lower extinction risk.

scaling results could in principle be reproduced by running enough small worlds and averaging over random seeds.

To test this explanation, we hold the total “sample size” approximately fixed while changing whether it is concentrated in large environments or partitioned across many smaller environments. In the **Beach** terrain with compass-equipped agents (**RC**), we consider three conditions:

- 4 runs of a 512×512 world with 8192 initial agents each,
- 16 runs of a 256×256 world with 2048 initial agents each,
- 64 runs of a 128×128 world with 512 initial agents each.

Across these conditions, the total number of grid cells, starting resources, and initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed runs for 2M steps with the same hyperparameters as Section 4.1. As a control, we repeat the protocol for agents without a compass (**R**). Table 10 summarizes long-distance mining and extinction frequency for each configuration.

If large environments were merely self-averages of many smaller ones, mining and extinction rates should be roughly invariant to how grid cells and agents are distributed across runs: 4 large worlds and 64 small worlds should show similar mining probabilities. Instead, we observe a strong dependence on world size. For (**RC**) agents, mining emerges in all runs with no extinctions at 512×512 , remains common but not guaranteed at 256×256 , and becomes unreliable and fragile at 128×128 , where many populations go extinct before reaching 2M steps. In contrast, (**R**) agents never develop mining and go extinct at all scales, confirming that the compass remains a key enabler of long-range behavior.

These results are inconsistent with a purely self-averaging account. Larger environments appear to enable and stabilize mining in ways that are not reproduced by averaging over more small simulations.

Table 10: Mining and extinction events when trading off world size against the number of independent runs. We compare **Beach** worlds at grid sizes 512×512 , 256×256 , 128×128 with initial populations scaled proportionally (8192, 2048, 512 agents, respectively). For each size, we allocate the same total number of grid cells and initial agents across all runs by running 4 seeds at 512×512 , 16 seeds at 256×256 , and 64 seeds at 128×128 . We report the fraction of runs in which mining emerged (🔍) and the fraction in which the population went extinct before 2M steps (💀), for agents with a compass (RC) and without a compass (R).

	512×512		256×256		128×128	
	8192		2048		512	
	🔍	💀	🔍	💀	🔍	💀
(RC)	4/4	0/4	15/16	1/16	31/64	41/64
(R)	0/4	4/4	0/16	16/16	0/64	64/64

Even with the same total number of cells, agents, and simulation steps, concentrating those resources into fewer large worlds yields a regime with consistent mining and rare extinctions, whereas distributing them across many small worlds yields populations that often fail to discover mining and frequently collapse. This suggests that environmental scale does more than smooth stochastic noise; it changes the ecological regime, for example by supporting larger spatially distributed populations and more persistent diversity among evolving policies.

F.4 MUTATION RATE SWEEP

We test how sensitive mining emergence is to the mutation rate, the hyperparameter controlling the scale of Gaussian noise added to a child’s policy at birth. Mutation rate directly controls how different offspring policies are from parent policies. We again use a 256×256 **Beach** world with an initial population of 2048 (RC) agents and a maximum length of 2M steps.

Figure 18 shows that for mutation rates from 6×10^{-3} to 7.875×10^{-2} , mining emerges in at least 3 of 4 seeds, indicating that the Section 4.1 results using mutation rate 3×10^{-2} are robust within this interval. As expected, mutation rates that are too small (e.g., 3×10^{-3}) prevent agents from developing new behavior and impair survival, leading to extinction. Conversely, large mutation rates (e.g., 3×10^{-1}) undermine population stability, consistently leading to extinction with no mining adaptation.

F.5 NETWORK SIZE

We study whether mining and extinction depend narrowly on the chosen policy-network size and whether environmental scale changes that dependence. We investigate 256×256 and 512×512 **Beach** worlds with initial population sizes of 2048 and 8192 (RC) agents, respectively (scaling by four with the number of grid cells). We ran baselines with (R) agents for each experiment, again finding rare to no mining events and 100% extinction rates.

The main experiments in Section 4 use policies with 2 layers and 64 channels, yielding 22465 parameters per agent (Table 11). In Figure 19 we show that in the 256×256 **Beach**, halving channels to 32 (resulting in 9185 total parameters) or doubling to 128 (giving 61313 parameters) with number of layers fixed has minimal to no effect on mining and extinction frequency. Meanwhile, changing network depth has a significant effect: all populations of agents with just 1 layer and 32 channels went extinct, and 50% of populations of agents with 3 layers and 128 channels went extinct.

Scaling the environment to 512×512 changes the sensitivity to network depth (Figure 20). In particular, in the larger world, agent populations with 3 layer, 128 channel networks consistently mined and survived through 2M steps. Surprisingly, agent populations with the small 1 layer, 32 channel networks mined consistently as well. We hypothesize that very small and very large networks face different discovery challenges. Smaller networks may lack capacity, while larger networks expand the search space and may require more time to find useful behavior. The fact that these networks often fail in small environments but succeed in larger ones illustrates that scale can make emer-

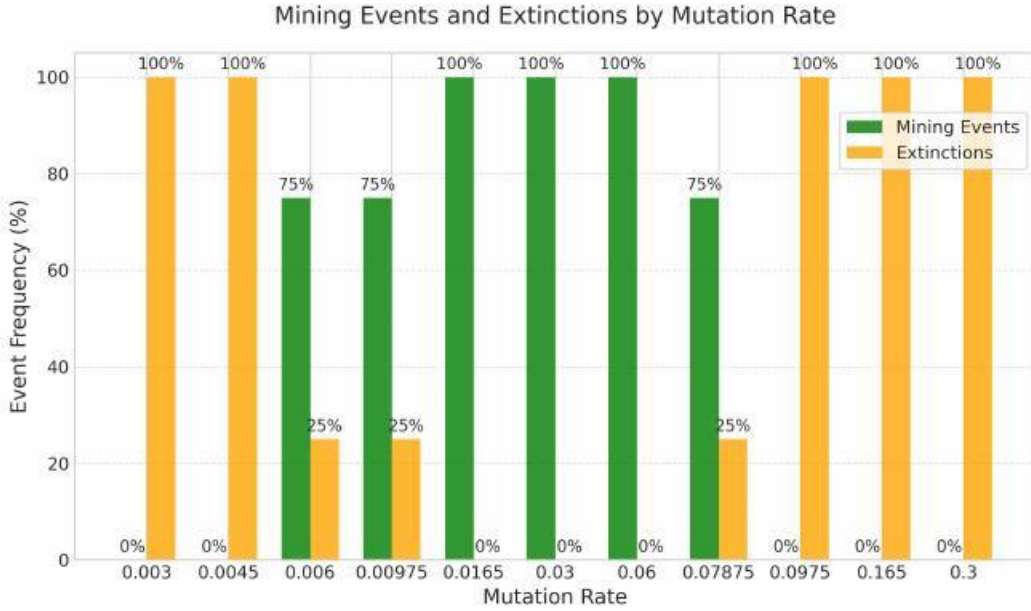


Figure 18: Frequency of mining and extinction events in a 256×256 **Beach** world across mutation rates. Frequencies are computed over four random seeds per setting.

gent behavior less dependent on a narrow architecture choice. Larger environments support larger populations, allowing broader exploration of policy space. At the same time, locally successful mutants take longer to propagate across a large environment, which can preserve diversity and further promote exploration. For related discussion, see [\(Mitchell et al. 2006\)](#).

Table 11: Total parameter counts for **(RC)** and **(R)** agent policies under each tested network architecture.

Architecture (layers, channels)	Parameter Count	
	(RC)	(R)
1, 32	8129	7969
2, 32	9185	9025
2, 64	22465	22145
2, 128	61313	60673
3, 128	77825	77185

F.6 VARYING VISION RANGE

We assess whether visual foraging and predation depend on a particular visual field by varying both vision range and viewing geometry in the 512×512 **Ocean** world with **(RCV+A)** agents. We evaluate visual sensor ranges of 3×3 , 7×7 , and 9×9 . For each range, we evaluate two viewing configurations shown in Figure 21: a) Around: agents can see grid cells all around them b) Front: agents cannot see cells behind them. The total number of cells in the agent’s vision range is equal for the two configurations. The main experiments with vision in Section 4.2 were run with the 7×7 Around setting.

These results show that a small 3×3 vision range produces behavior close to the no-vision (Blind) setting, whereas 9×9 behaves similarly to 7×7 . This indicates that there may be diminishing returns of larger vision ranges in the current environmental configuration. This is intuitive given the rest of the environmental configuration; since the agents have no memory, it is unlikely that visual

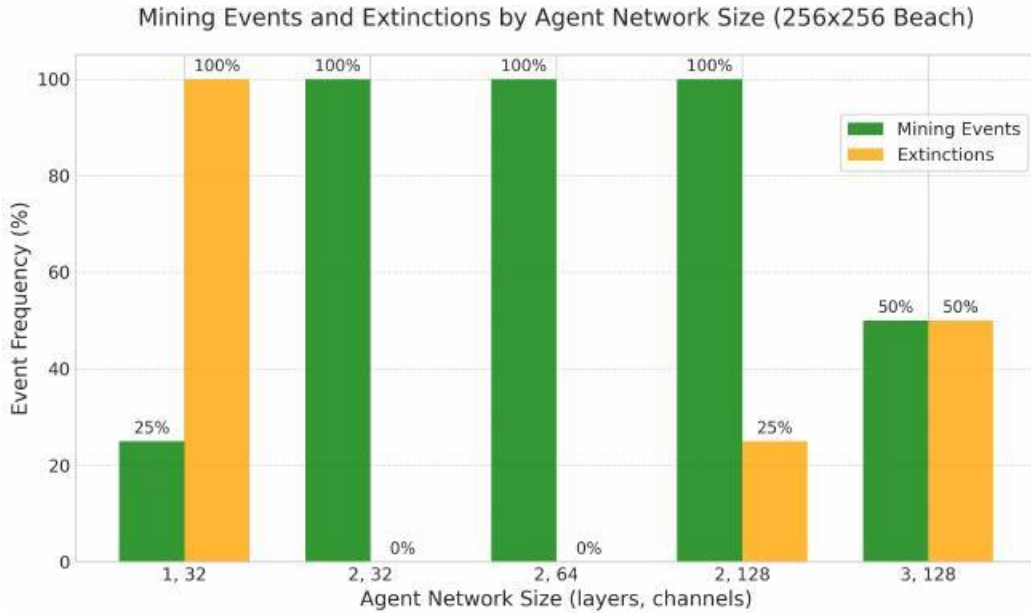


Figure 19: Frequency of mining and extinction events in a 256×256 **Beach** world across agent policy network sizes. Frequencies are computed over four random seeds per setting.

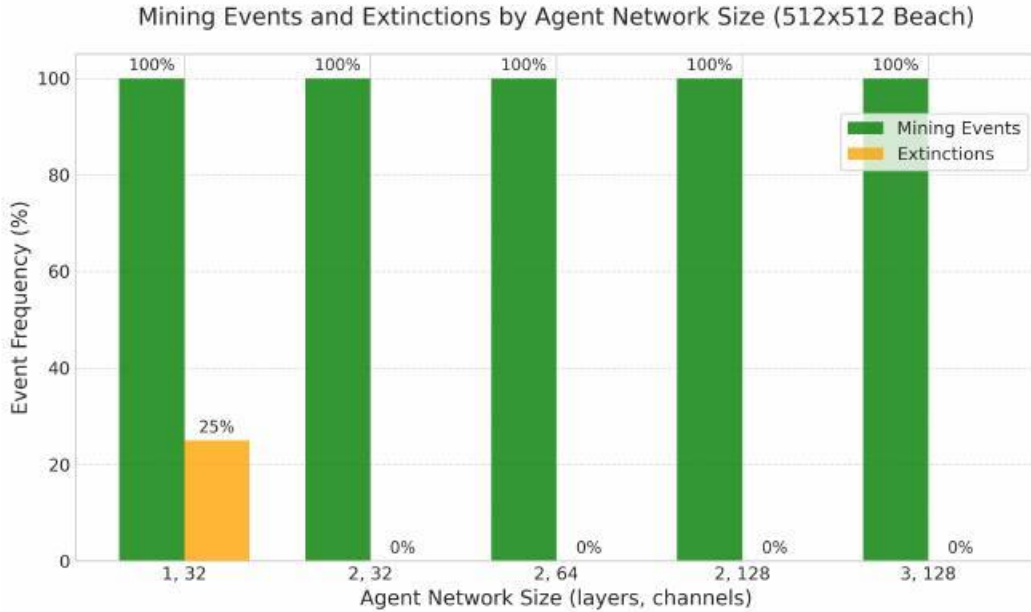


Figure 20: Frequency of mining and extinction events in a 512×512 **Beach** world across agent policy network sizes. Frequencies are computed over four random seeds per setting.

features at the periphery of a 9×9 window could meaningfully alter the optimal action relative to a 7×7 window consistently enough to provide a strong learning signal.

The data also suggest that at 7×7 and 9×9 ranges, the Around configuration is slightly more favorable for population stability and for effective attack and eat actions, resulting in higher average population 🧑, biomass utilization ⚙️, and homicides per attack 🎯 than the Front configuration. We hypothesize that this is due to diminishing returns of being able to see more distant pixels; in

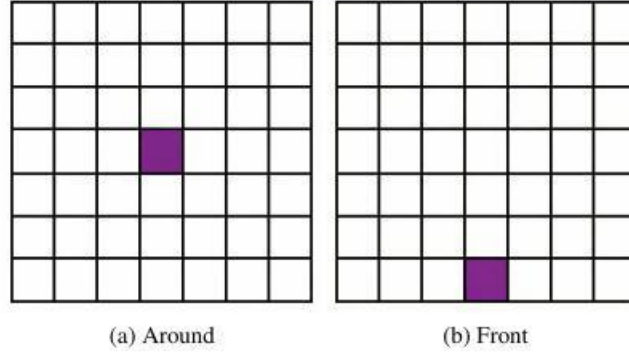


Figure 21: Comparison of the Around and Front viewing configurations for the 7×7 vision range. The purple cell marks an agent facing toward the top of the page.

Table 12: Foraging and predation data in the 512×512 **Ocean** world for vision agents (**RCV+A**) with varying vision ranges and viewing configurations. Refer to Tables 5 and 6 for definitions of the metrics represented by each icon. For each seed, the value for each metric was averaged across all steps starting at step 500k to allow them to stabilize. Reported values are the mean across the four seeds and the standard deviation of the four seed averages.

Vision Range	Viewing Configuration						
3×3	Blind	7212 ± 2496	$.181 \pm .060$	$.435 \pm .090$	$.392 \pm .033$	$.032 \pm .006$	$.307 \pm .090$
	Around	10308 ± 4016	$.258 \pm .105$	$.352 \pm .014$	$.427 \pm .022$	$.040 \pm .030$	$.388 \pm .195$
	Front	13863 ± 520	$.351 \pm .023$	$.336 \pm .008$	$.437 \pm .009$	$.026 \pm .006$	$.470 \pm .145$
7×7	Around	16739 ± 122	$.373 \pm .003$	$.259 \pm .000$	$.532 \pm .001$	$.015 \pm .000$	$.751 \pm .007$
	Front	15255 ± 623	$.343 \pm .015$	$.253 \pm .001$	$.540 \pm .004$	$.015 \pm .000$	$.690 \pm .041$
9×9	Around	16744 ± 117	$.372 \pm .002$	$.257 \pm .001$	$.534 \pm .001$	$.015 \pm .000$	$.728 \pm .009$
	Front	15211 ± 1416	$.342 \pm .033$	$.253 \pm .001$	$.541 \pm .007$	$.015 \pm .000$	$.664 \pm .050$

the Around configuration, more pixels in the agent’s vision range are nearby, likely containing more relevant information for near-term actions than distant cells visible with the Front configuration. At the 3×3 vision range, however, the trend is reversed, with the Front configuration being more favorable than Around; this indicates that in this environment, the marginal benefit of seeing one more pixel ahead through the 3×3 Front configuration outweighs that of seeing one pixel behind in the 3×3 Around configuration, which gives most similar behavior to the Blind setting.

F.7 VARYING ATTACK STRENGTH

We study whether predation and population dynamics depend on the lethality of the attack action in the 256×256 **Ocean** world. We evaluate the following attack strengths: 1) Full Attack: deals 10 HP of damage, instantly killing any agent in the attack radius 2) Half Attack: deals 5 HP of damage 3) Quarter Attack: deals 2.5 HP of damage. The main **Ocean** experiments in Section 4.2 were run with the Full Attack setting.

Reducing attack strength from Full Attack to Half Attack substantially reduces both attack effectiveness and attack frequency: attacks per agent and homicides per attack drop considerably (Figure 22). As a result, populations with lower attack strength grow larger, with agents moving and eating more frequently, resulting in greater biomass utilization. Further reducing strength to the Quarter Attack setting yields diminishing returns in terms of further reduction in violence and facilitation of additional population growth. Intuitively, when attacks do less damage, a homicide requires more attacks per agent, reducing the lethality of each attack. However when the lethality is reduced, attacking behavior becomes less useful and therefore less likely to be selected for in the agent population. Agents with lower attack strengths therefore do not become as skillful at attacking and spend more time foraging for food.

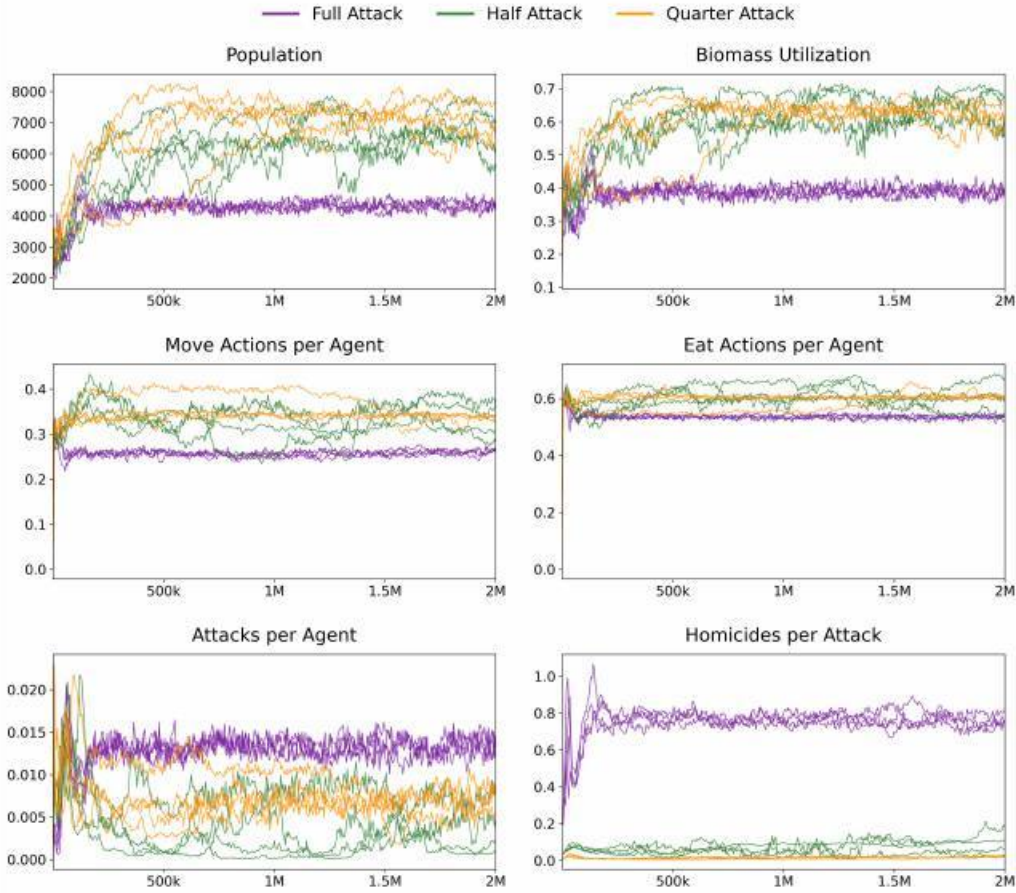


Figure 22: Population and action-frequency trajectories over 2M steps in a 256×256 **Ocean** world for (RCV+A) agents under Full Attack, Half Attack, and Quarter Attack settings. Four seeds are plotted per setting.

F.8 VARYING AGENT HP

We investigate whether adding stronger lifetime growth pressure changes the observed behaviors by reducing child starting HP and lowering healing rate. The healing rate parameter determines how much HP an agent can regenerate per healing step, which converts available energy to HP. Agents can influence healing only by maintaining enough energy.

We study two settings: 1) Full HP: child HP starts at 5 and healing rate is 1 HP per 0.1 energy 2) Low HP: child HP starts at 1, with healing rate 0.5. Max agent HP is 10 in all settings. The main experiments (Section 4) were run with the Full HP setting.

Across the two settings, homicides per attack, eat action frequency, and biomass utilization are not substantially different (Figure 23). In the Full HP setting, populations grow slightly larger, while in the Low HP setting, agents attack less frequently and move slightly more. We hypothesize that agents born with lower HP must focus on foraging early on in their lifetime in order to grow to full size, which would explain this difference.

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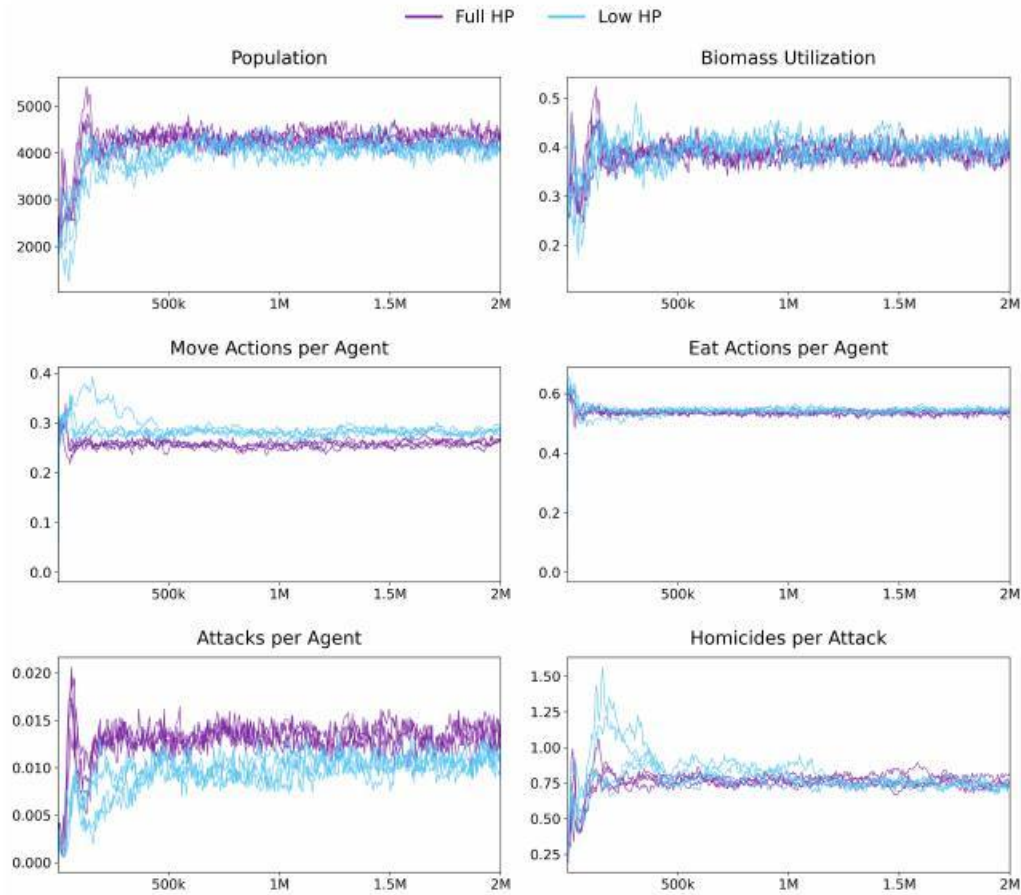


Figure 23: Population and action-frequency trajectories over 2M steps in the 256×256 **Ocean** world for (RCV+A) agents under Full HP and Low HP settings. Four seeds are plotted per HP setting.